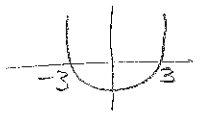


### Question 1

•  $D_f: t^2 - 9 > 0$    
 So  $D_f = \{x \mid x < -3 \text{ or } x > 3\}$   
 $= (-\infty, -3] \cup [3, \infty)$

• Critical numbers:

$f'(t) = 0 \Rightarrow t = 0 \notin D_f$

$f'(t)$  does not exist if

$t = \pm 3 \in D_f$

Only critical numbers is

$t = \pm 3$

3.2  $f(x) = \frac{\ln x}{x^2}$   $D_f = (0, \infty)$

$f'(x) = \frac{(\frac{1}{x})(x^2) - (\ln x)(2x)}{(x^2)^2}$   
 $= \frac{x(1 - 2\ln x)}{x^4} = \frac{1 - 2\ln x}{x^3}$   $(x > 0)$

Critical number:

•  $f'(x) = 0 \Rightarrow 1 - 2\ln x = 0$   
 $\Rightarrow \ln x = \frac{1}{2}$   
 $\Rightarrow x = e^{\frac{1}{2}} = \sqrt{e}$

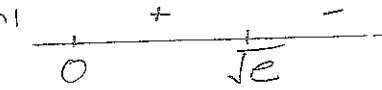
$f$  is increasing if

$1 - 2\ln x > 0$

$\Rightarrow 2\ln x < 1 \Rightarrow \ln x < \frac{1}{2}$

$\Rightarrow x < \sqrt{e}$

But  $D_f = (0, \infty)$   
 $\therefore f$  is increasing on  $(0, \sqrt{e})$

3.3  $f'$  

$f$  has a local maximum at  $x = \sqrt{e}$  which is  $f(\sqrt{e}) = \frac{\ln e^{\frac{1}{2}}}{(e^{\frac{1}{2}})^2} = \frac{\frac{1}{2} \ln e}{e} = \frac{1}{2e}$

### Question 2

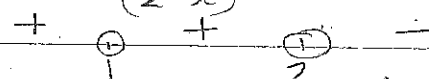
$f(x) = \frac{x - x^2}{x^2 - 3x + 2} = \frac{x(1-x)}{(x-1)(x-2)}$

2.1  $D_f = \mathbb{R} \setminus \{1, 2\} = \frac{-x}{x-2} = \frac{x}{2-x}$

$f'(x) = \frac{(2-x) - (x)(-1)}{(2-x)^2} = \frac{2}{(2-x)^2}$

$f''(x) = \frac{0(2-x)^2 - (2)(2)(2-x)(-1)}{(2-x)^4}$

$= \frac{4}{(2-x)^3}$

$f''$  

$f$  CU on  $(-\infty, 1)$  and on  $(1, 2)$

$f$  CD on  $(2, \infty)$

2.2 No inflection point

since  $2 \notin D_f$

(at  $x=2$   $f''$  changes from  $+$  to  $-$ )

### Question 3

3.1 Let  $x_1, x_2 \in I$

$f$  is increasing on  $I$  if  $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$

### Question 4 $D_f = [-1, 0]$

$f(x) = xe^{-x}$   $f$  continuous on  $[-1, 0]$

•  $f'(x) = e^{-x} - xe^{-x}$   
 $= e^{-x}(1-x)$   
 $\begin{matrix} > 0 & = 0 \end{matrix}$

Critical number:

$f'(x) = 0 \Rightarrow x = 1 \notin D_f$

No critical number

|        |      |   |
|--------|------|---|
| $x$    | -1   | 0 |
| $f(x)$ | $-e$ | 0 |

Absolute max is  $f(0) = 0$

Absolute min is  $f(-1) = -e$